

Can the model drill Meier's granite block at the measured speed?

A pilot-scale flame-jet drilling experiment from an ETH Zürich thesis, rebuilt as a simulation, and what the first complete runs show.

Status as of 17 September 2026. Written for a reader outside the project; no knowledge of the code is assumed.

In one paragraph. In 2017 Thierry Meier drilled through a 0.5 m thick, 660 kg block of Grimsel granite with a 38 kW methane–air flame and no mechanical cutting: the hot jet makes the rock surface flake away, and the burner, standing on short feet, follows the rock down. He recorded how fast it sank (about 1.5 m per hour; 1.3–1.6 m/h once his numbers are made consistent with each other), the hole diameter (85 mm) and the volume removed (3.42 litres, by filling the hole with water). We simulated this with a three-dimensional heat-conduction and rock-flaking model. All inputs come from the experiment, from the burner drawings or from published correlations. The two that were never measured, the gas temperature at the nozzle and how strongly the jet heats the rock, were set to their most defensible values before any run, and nothing was adjusted afterwards.

The model sinks at 1.36 m/h, inside the measured range, and its own removal rates put the hole below the first 10 cm at 85–92 mm wide. Two things are wrong. The model flares the top of the hole into a 200 mm funnel that the sectioned block does not show (Figure 6), so it removes about twice the measured volume; and the deep centre of the hole never settles, so the runs are transients of a few minutes rather than the experiment's steady 23 minutes. The agreement also rests on the nozzle gas temperature, which was never measured: the chamber thermocouple sat upstream of the lifted flame and read about 550 °C during drilling, so the adiabatic 1900 K used here is an upper bound, and a 1436 K flame drills at half the measured rate. The missing or over-stretched physics behind each of these is listed at the end.

1. The experiment

Source. T. Meier, *Thermal spallation drilling*, ETH Zürich, Diss. No. 24021 (2017), Chapter 8, pp. 193–226: the pilot-scale demonstration on a 660 kg block. Burner dimensions come from the burner drawing set (VT5, sheets 1, 14, 15) that accompanies the thesis work.

What was done. A monolithic block of Grimsel granite, $0.7 \times 0.7 \times 0.5$ m, was clamped lightly from the sides (about 1 MPa, only to stop it splitting). A water-cooled burner burning 2.47 kg/h of methane in 52 kg/h of air, 38 kW of chemical power, hung from a crane on coiled tubing above the block. Its hot exhaust left through a small nozzle at the bottom and hit the rock. Heated quickly, the granite surface expands against the cold rock beneath it and thin flakes pop off; this is *thermal spallation*. As the rock under the burner disappears, the burner sinks. It rests on three short feet, so it can only sink as fast as the rock under those feet is removed. In 1383 s of actual drilling (two phases with a pause in between) it went through the whole 0.5 m block.

How the run went (thesis pp. 218–221 and Figure 2). After ignition the methane and air were ramped up over about ten minutes to their set points of 2.47 and 52 kg/h. Drilling began only after that ramp, once both flows were steady, and lasted 647 s in the first phase and 736 s in the second. Between the two phases a vacuum cleaner was fitted to the exhaust line, which reversed the flow in the gap between burner and wellhead; in the second phase the air flow also dipped briefly at 1250 s. Neither changed the rate of penetration, which the thesis reports as constant at 1.5 m/h over both phases. The simulation therefore represents the start of a drilling phase at full, steady flow; the ramp is not modelled, and nothing in the model distinguishes phase 1 from phase 2. The thermocouple curve in Figure 2 is discussed under the table below.

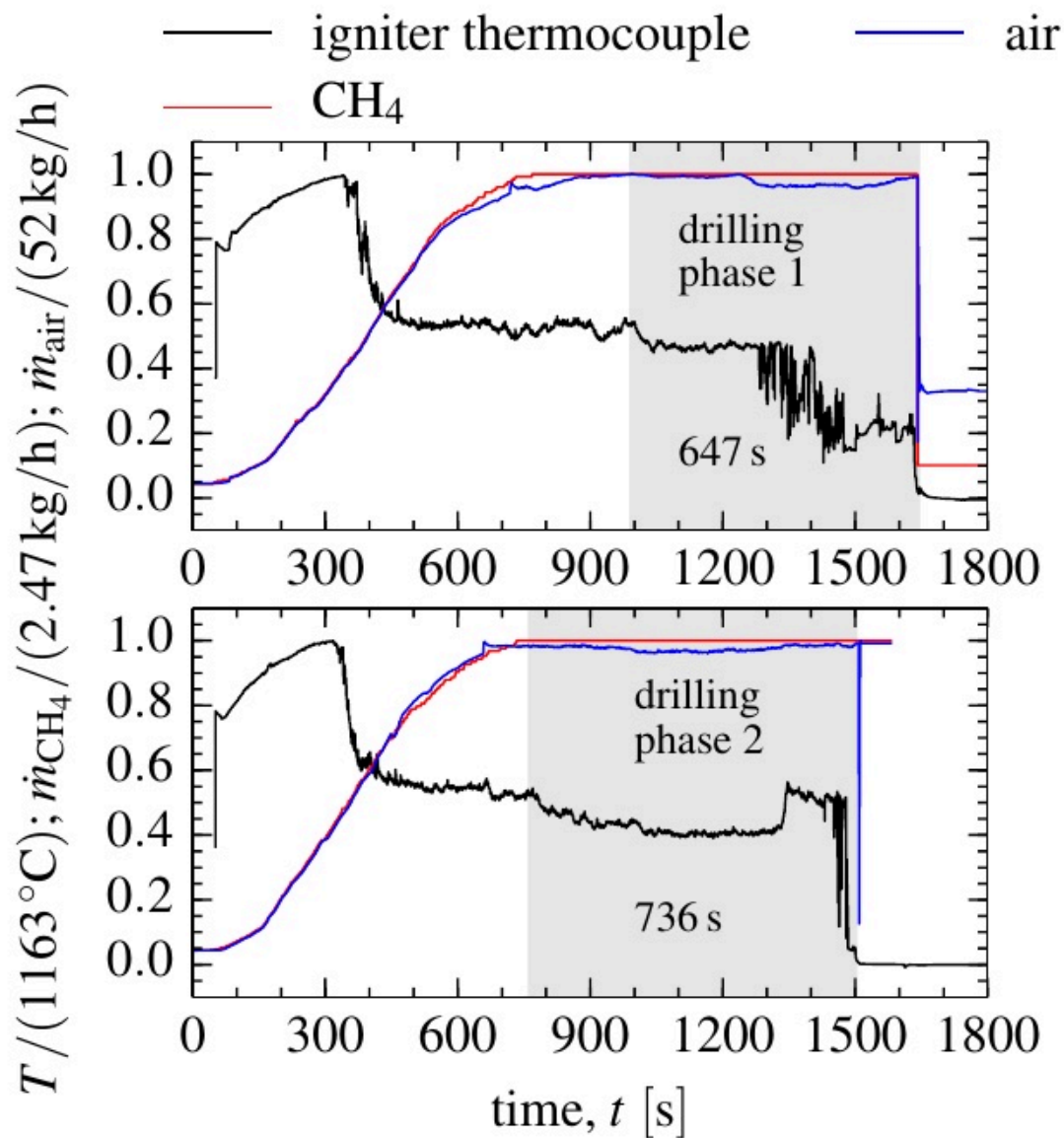


Figure 2. Meier's own record of the run (his Figure 8.7, p. 220). Each curve is divided by its set point, so all three read 1.0 at their maximum: methane by 2.47 kg/h, air by 52 kg/h, and the igniter thermocouple by 1163 °C. The shaded windows are the two drilling phases. The thermocouple peaks at 1163 °C early in the ramp, while the flame is still attached to the injector, then drops to about half that when the flame lifts off, and stays near 0.4–0.5 for the whole of both drilling phases. This sensor is the ignition coil's own thermocouple, 33 mm from the fuel injector; once the flame has lifted it sits upstream of the flame front and does not see the combustion products that leave the nozzle.

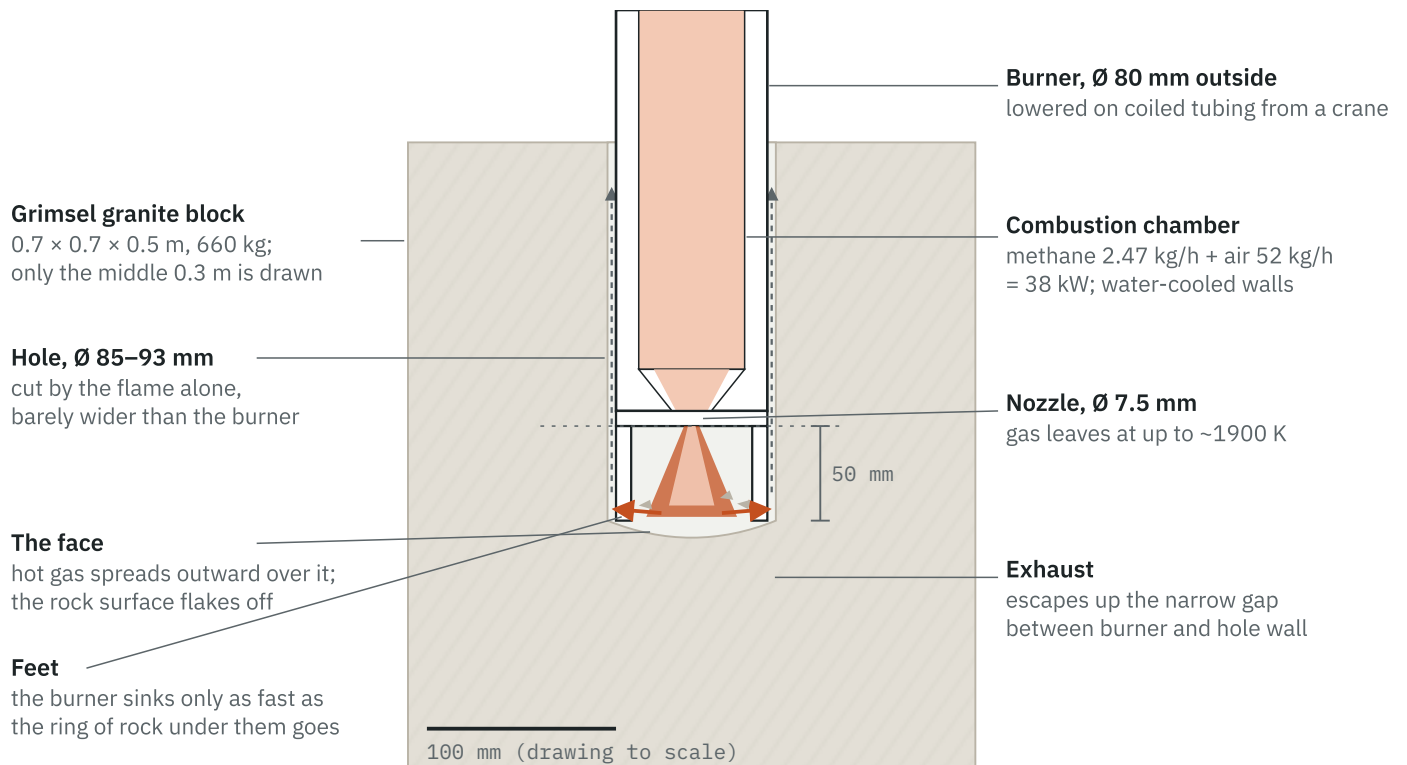


Figure 1. How the experiment drills. Cross-section through the burner and the block. The hot jet strikes the bottom of the hole, spreads outward over it, and makes the surface flake away. The burner stands on three feet at the edge of that face, so its rate of descent is the rate at which the outer ring of the face is removed. The nozzle sits 50 mm above the feet. Dimensions from the burner drawings; the block is 0.7 m wide, of which the central 0.3 m is shown.

What was measured, and what we can actually score

QUANTITY	REPORTED	OUR TARGET	WHY THE TARGET DIFFERS
Rate of penetration (how fast the burner sank)	1.5 m/h	1.3–1.6 m/h	1.5 m/h for 1383 s would be 0.58 m, but the block is 0.50 m. The consistent range is 1.3–1.6 m/h. We accept $\pm 20\%$ around it: 1.04–1.92 m/h.
Hole diameter	85 mm	85–93 mm, never below 80	Thesis Table 8.2, p. 222, and the section photograph (Figure 6). 3.42 L over 0.50 m implies a mean diameter of 93 mm. The burner itself is 80 mm, so the hole cannot be narrower (p. 223).
Volume removed	3.42 L in 1383 s	2.0–3.0 cm ³ /s	Measured by filling the finished hole with water (p. 221), so it is the whole excavation including any widening at the mouth. The model's whole excavation is compared with it.
Gas temperature at the nozzle	not measured	1900 K used; 1436 K as a low case	The only temperature sensor is the igniter thermocouple in the flame's recirculation zone (p. 212). Its reading is reported uncorrected (p. 215), was calibrated only to 1000 °C, peaked at 1163 °C before the flame lifted off, and read about 550 °C throughout drilling (Figure 2). None of these is the product-gas temperature. The adiabatic flame temperature of this mixture, about 1900 K, is

QUANTITY	REPORTED	OUR TARGET	WHY THE TARGET DIFFERS
			an upper bound; heat lost to the burner's cooling water, which the chapter does not quantify, lowers it. 1436 K (1163 °C) is kept as a low case, not as a measurement. The burner's cooling water ran at 160 L/h with thermocouples on inlet and outlet (p. 208), but the temperature rise is not reported; each 10 K of rise would be 1.9 kW and lower the nozzle gas by about 100 K.
Rock heating before a flake leaves	not measured	500–560 K rise	Derived from the removal energy the volume rate implies. A model consistency check rather than a measurement.

Two things were *not* measured and matter most: how effectively the hot gas hands its heat to the rock (the heat transfer coefficient), and the gas temperature at the nozzle exit. Those are the model's two genuine unknowns.

2. How the experiment became a simulation

The simulation is written in ALAMO, a block-structured finite-volume code. Six pieces of physics are chained together. Each is given here with the equation the code actually evaluates and the values used.

2.1 Heat conduction in the rock

The granite is a grid of 2 mm cubes with the thesis's measured properties. Temperature obeys the heat equation, advanced explicitly with a 4 ms step:

$$\rho c_p \partial T / \partial t = \nabla \cdot (k \nabla T)$$

$$k = 1.5 \text{ W/m K}, \rho = 2750 \text{ kg/m}^3, c_p = 790 \text{ J/kg K}, \alpha = k / (\rho c_p) = 6.9 \times 10^{-7} \text{ m}^2/\text{s}$$

The block starts at 293 K. By symmetry one quarter of the hole is computed: a box $0.12 \times 0.12 \text{ m}$ across and 0.30 m deep ($60 \times 60 \times 150$ cells), with the jet axis on the corner edge. All outer faces are insulated; the two faces through the axis are symmetry planes.

2.2 Why and when a flake leaves

When the surface is heated, its thermal expansion is restrained by the cold rock beneath, which puts the heated skin under an in-plane stress. For a layer at temperature T above the initial T_0 ,

$$\sigma(z) = \frac{E \alpha_{\text{th}} (T(z) - T_0)}{1 - \nu}, \quad E = 30 \text{ GPa}, \alpha_{\text{th}} = 8 \times 10^{-6} \text{ K}^{-1}, \nu = 0.3$$

Granite contains microcracks. The model follows the spallation literature in taking the magnitude of this stress as the load on a crack of length a running parallel to the surface, and evaluates the stress-intensity factor with the standard edge-crack weight function, integrating the stress over the crack's depth:

$$K_I = \frac{2}{\sqrt{\pi a}} \int_0^a \sigma(z) \frac{g(z/a)}{\sqrt{(1 - (z/a)^2)}} dz, \quad g(\xi) = 1.3 - 0.3 \xi^{5/4}$$

A surface cell is removed the first time K_I reaches the fracture toughness $K_{\text{Ic}}(T)$ of granite, which falls with temperature (Nasseri 2007, Westerly granite: $1.43 \text{ MPa}\sqrt{\text{m}}$ at room temperature, 0.98 at 450°C , 0.43 at 650°C). The crack length is not a fitted number. Each cell draws its own flaw from a Weibull distribution,

$$a = a_0 (-\ln U)^{1/m}, \quad U \text{ uniform on } (0, 1), \quad m = 20$$

$$a_0 = 4 \text{ }\mu\text{m} \text{ inside grains and } 20 \text{ }\mu\text{m} \text{ on grain boundaries}$$

on a Voronoi grain structure of about 2 mm grain size. With these values the criterion is met when the surface has warmed by about 530 K , to a *firing temperature* of about 820 K ; the run-to-run spread is a few kelvin. Check: at

528 K of heating the stress is 181 MPa, which on a 4.4 μm crack gives $K_I \approx 0.74 \text{ MPa}\sqrt{\text{m}}$, against $K_{Ic} \approx 0.71 \text{ MPa}\sqrt{\text{m}}$ at 820 K. When a cell fires, the surface recedes by half a cell (1 mm) and the cell below becomes the new surface. Flake size is therefore set by the grid, not predicted; what the grid does not affect is the energy each removed volume carries away, $\rho c_p \Delta T_{\text{fire}} \approx 1.15 \text{ GJ/m}^3$. For comparison, Rauenzahn's compilation of measured surface-temperature rises at spallation for Barre and Westerly granite spans roughly 300–900 °C over 0.6–1.8 MW/m² (thesis Fig. 2.12, p. 49), and the Weibull exponent measured for granite is 15–25 (p. 48); the model's 530 K at 0.4–1.0 MW/m² and $m = 20$ sit inside both.

2.3 The heat flux on the rock surface

Once a column of rock has fired, its surface stays at the firing temperature while it keeps flaking. The heat it receives per unit area is then the gas heating minus radiation and free convection to the surroundings:

$$q = h(r, s) (T_{\text{gas}} - T_{\text{fire}}) - \varepsilon \sigma_{\text{SB}} (T_{\text{fire}}^4 - T_0^4) - h_c (T_{\text{fire}} - T_0)$$

$$\varepsilon = 0.8, h_c = 10 \text{ W/m}^2 \text{ K}$$

and the removal rate of that column follows from energy conservation alone:

$$v = \frac{q}{\rho c_p (T_{\text{fire}} - T_0)}$$

Columns that have never fired are heated as ordinary conducting rock until they do. $s = z_{\text{nozzle}} - z_{\text{face}}$ is the height of the nozzle above that column's surface, and r its distance from the jet axis. A column whose surface is above the nozzle plane ($s \leq 0$) is outside the jet and receives nothing.

2.4 How strongly the jet heats the rock: Martin's correlation

The heat transfer coefficient $h(r, s)$ of a round jet striking a flat wall was correlated by Martin (1977) from many experiments. His formula gives the Nusselt number averaged over a disc of radius r around the axis, for a nozzle of diameter D at height H :

$$\frac{\bar{h}(r) D}{k_{\text{gas}}} = \text{Pr}^{0.42} \cdot F(\text{Re}) \cdot G(r/D, H/D)$$

$$F = 2 \text{Re}^{1/2} (1 + 0.005 \text{Re}^{0.55})^{1/2}, \quad G = \frac{(D/r) (1 - 1.1 D/r)}{1 + 0.1 (H/D - 6) D/r}$$

The model needs the *local* coefficient at each radius, which follows from the averaged one by differentiating the heat collected inside radius r , $h = d(r^2 \bar{h})/d(r^2)$. Writing $c = 0.1 (H/D - 6)$, this gives

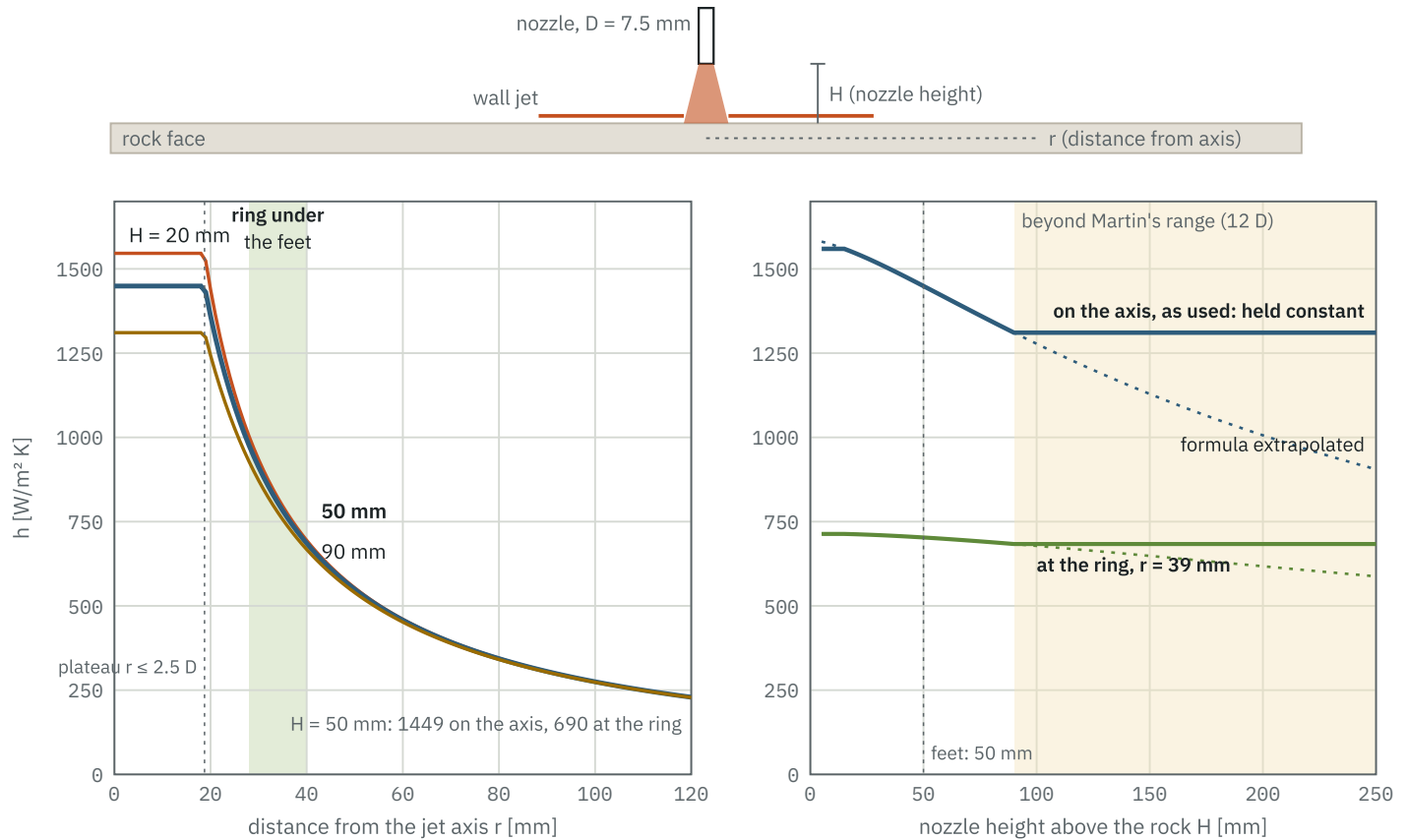
$$h(r, H) = \frac{k_{\text{gas}}}{D} \text{Pr}^{0.42} F(\text{Re}) \cdot \frac{D}{2r} \cdot \frac{r^2 + 2c D r - 1.1 c D^2}{(r + c D)^2}$$

Validity as stated by Martin: $2000 \leq \text{Re} \leq 400\,000$, $2.5 \leq r/D \leq 7.5$, $2 \leq H/D \leq 12$. Inside $r = 2.5 D$ the model holds the value at $2.5 D$ (a flat stagnation plateau); H is clamped to $2-12 D$.

Everything in it is fixed by the experiment. With $D = 7.5$ mm from the drawings and the measured mass flow of 0.01513 kg/s, the jet Reynolds number $\text{Re} = 4\dot{m}/(\pi D \mu)$ is $51\,600$, with air properties at the film temperature $(1900 + 820)/2 = 1360$ K ($k_{\text{gas}} = 0.080$ W/m K, $\text{Pr} = 0.73$, $F = 781$). At the nozzle height of the feet, $H = 50$ mm $= 6.7 D$, the coefficient under the jet is 1449 W/m² K, and at the ring under the feet ($r = 39$ mm) it is 690 W/m² K. Figure 3 shows the shape. The two bracket values used in earlier work, 5000 and $10\,000$ W/m² K, came from the literature rather than from this jet: Meier's own Chapter 7 model used $10\,000$ W/m² K, a figure taken from Potter Drilling, and the project's reference sheet gave $5\,000-30\,000$ as a plausible band. They are kept as a bracket and turn out to drill too fast (§3).

Check of the numbers. The inputs were re-derived independently for this report: with the film-temperature air properties (viscosity 5.0×10^{-5} Pa s) the Reynolds number is $51\,640$, the Prandtl number 0.726 , $F = 781$, and $G(2.5 D, 6.7 D) = 0.218$, giving an averaged 1600 and a local 1449 W/m² K; the study's values agree. Two caveats belong here. Evaluating the properties at the nozzle temperature instead of the film temperature would give 1552 W/m² K, at the firing temperature 1295 , so the property convention alone is worth $\pm 7\%$. More seriously, the burner's nozzle is a converging-diverging Laval nozzle with a 6 mm throat (pp. 212, 222); at this mass flow that throat is choked, the chamber pressure, which the thesis says is set by the 6 bar gauge air supply (pp. 208, 211), is near 6 bar absolute, and the jet leaves supersonic. Martin's data are for subsonic jets. The Reynolds number is defined by the mass flow and is unaffected, but the structure of a

supersonic impinging jet, with shocks near the wall, is not what the correlation describes. This is listed among the limitations. One more point of reference: the thesis's own impinging hot-air jets in its sensor-calibration rig (5.3 mm nozzle, 400–700 °C, Reynolds numbers of a few times 10^4 , nozzle heights of 2 to 6 diameters) gave measured coefficients of 0.4–1.6 kW/m² K (Fig. 3.12, p. 91). That is the same order as the correlation value here and well below the 5 000–10 000 W/m² K used earlier, which appear in the thesis only as an assumption borrowed from Potter Drilling for his Chapter 7 model (p. 193).



Left panel: — nozzle 20 mm above the rock — 50 mm (the feet height, used in the runs) — 90 mm

■ ring of rock under the feet, 28–40 mm

Right panel: — on the jet axis, as the model uses it — on the axis, Martin's formula continued past its range

— at the ring under the feet ■ nozzle heights beyond Martin's range (over $12 D = 90 \text{ mm}$)

Figure 3. How strongly the jet heats the rock, from Martin's correlation at the experiment's flow. Top: the geometry, with H the nozzle height above the rock and r the distance from the jet axis along the rock. **Left panel, moving outward along the rock face:** the coefficient is flat under the jet, falls by half between the axis and the ring under the feet, and by a further half by 80 mm. Raising the nozzle from 20 to 90 mm lowers the coefficient under the jet by 15 % and changes almost nothing at the ring; that is why the three curves lie on top of each other beyond 40 mm. **Right panel, raising the nozzle:** the same two positions, on the axis and at the ring, plotted against nozzle height. Martin measured up to 12 diameters (90 mm). Beyond that the model holds the value constant (solid line), whereas the formula continued would keep falling (dotted). Because of this clamp the centre of the hole, once it is deeper than 90 mm below the nozzle, keeps almost twice the ring's heating however deep it goes, which is what lets it run ahead without limit (§4).

2.5 The gas cools as it spreads, and the hole recirculates its exhaust

Martin's correlation gives the coefficient but not the gas temperature, and a fixed gas temperature would let the rock take more heat than the flame delivers. The model therefore carries the jet's heat budget explicitly. Free jets keep their nozzle temperature only within a core of about five diameters; beyond it they mix with the surrounding gas. The temperature and mass flow arriving at the centre of the face at height s_c are taken as

$$\varphi = \min(1, 5D/s_c), \quad T_{\text{stag}} = T_{\text{mix}} + (T_{\text{nozzle}} - T_{\text{mix}}) \varphi, \quad m = \dot{m}/\varphi$$

where T_{mix} is the temperature of the gas the jet entrains. In a blind hole the only gas available is the jet's own exhaust flowing back up, so T_{mix} is set to the previous time step's exhaust temperature (starting from 293 K). The gas then spreads outward over the face as a wall jet, and its temperature is marched outward in 2 mm radial rings, debiting the heat given to every rock column in the ring:

$$m c_{p,\text{gas}} [T_{\text{gas}}(r + \Delta r) - T_{\text{gas}}(r)] = - \sum_{\text{columns in ring}} \max(0, q) \Delta x \Delta y$$

$$c_{p,\text{gas}} = 1250 \text{ J/kg K}$$

starting from T_{stag} on the axis. The march ends where the gas runs out of face; the temperature there is the exhaust temperature. Every step the code checks that the heat delivered to the rock plus the heat leaving in the exhaust equals the heat that entered, $\dot{m} c_p (T_{\text{nozzle}} - T_{\text{mix}})$ plus the recirculated part, to round-off. The nozzle temperature is 1900 K in the main run (adiabatic flame of this mixture) and 1436 K in a sensitivity run (chamber thermocouple reading).

2.6 The burner follows the rock down

In the experiment the burner hung from the crane and was lowered so that it rested on the rock; the hook load showed whether it hung or lay on the rock (p. 221), and the drawworks gave it small intermittent pushes "to keep the drill free" (p. 219). In the model it is simply set down on its feet each step and never pushed. The thesis says only that "burner feet are used to keep a distance between the nozzle and the exposed rock surface" and that their height gives a stand-off of seven nozzle diameters, about 50 mm (p. 212, with the 7.1 mm nozzle quoted there). The burner drawings (assembly sheet 1/17, section A-A; holder sheet 15/17) show what the feet are: the 80 mm outer tube continues 50 mm below the nozzle flange, with a 56 mm bore and three slots cut through the wall for the exhaust. So the feet are neither three legs nor a closed ring but a slotted ring: three arc-shaped pads on the annulus between 28 and 40 mm from the axis. In the model that annulus is split into three sectors, one per pad. Each time step the height of the rock under each foot is taken as the 90th percentile of the surface heights in that foot's sector of the ring, the burner is set on the mean of the three, and the nozzle is placed 50 mm above:

$$z_{\text{feet}} = \frac{1}{3} \sum_{\text{pads}} Q_{0.9}\{z_{\text{face}} : 28 \text{ mm} \leq r \leq 40 \text{ mm, in pad}\}$$

$$z_{\text{nozzle}} = \min(z_{\text{nozzle}}^{\text{prev}}, z_{\text{feet}} + 50 \text{ mm})$$

The burner never rises. If the ring stops receding, the burner stops: that is recorded as a stall, and it is what the room-air run in §3 does. The rate of penetration is the descent rate of z_{nozzle} .

2.7 What was fixed in advance

All of the above was set from the experiment, the drawings and the literature before any run. A hand calculation of the expected rate and hole profile was written down first, then the runs were made and scored as they came out. The two unmeasured inputs, the nozzle temperature and the heat transfer coefficient, were run at their most defensible values (adiabatic flame; Martin at the measured flow) and bracketed by the alternatives. No parameter was adjusted toward the measured values. Simulating 340 s of drilling takes about 40 min on four processor cores; the main run was repeated with 1 mm cells.

On the heat transfer coefficient in particular. Earlier work in this project used 5 000 and 10 000 W/m² K, taken from the literature (Meier's Chapter 7 model, Potter Drilling) because nothing better was at hand. When the burner-on-feet model was specified, the correlation value at the measured flow, 1449 W/m² K, was written into the specification as the one to be scored, on the ground that it is the only value derived from this jet's own flow and geometry; the two literature values were kept as a bracket. That decision was taken before any burner-on-feet run existed. It was informed by earlier one-dimensional estimates, which had shown that the low end of the literature band gave roughly the right volume removal rate, but the number 1449 itself was never adjusted, and no run with any other value of h was used to pick it. The sensitivity in §3 shows how easy a fit would have been: the rate scales as roughly $h^{0.45}$, so almost any target could have been hit by choosing h . That is exactly why the value had to come from outside the comparison.

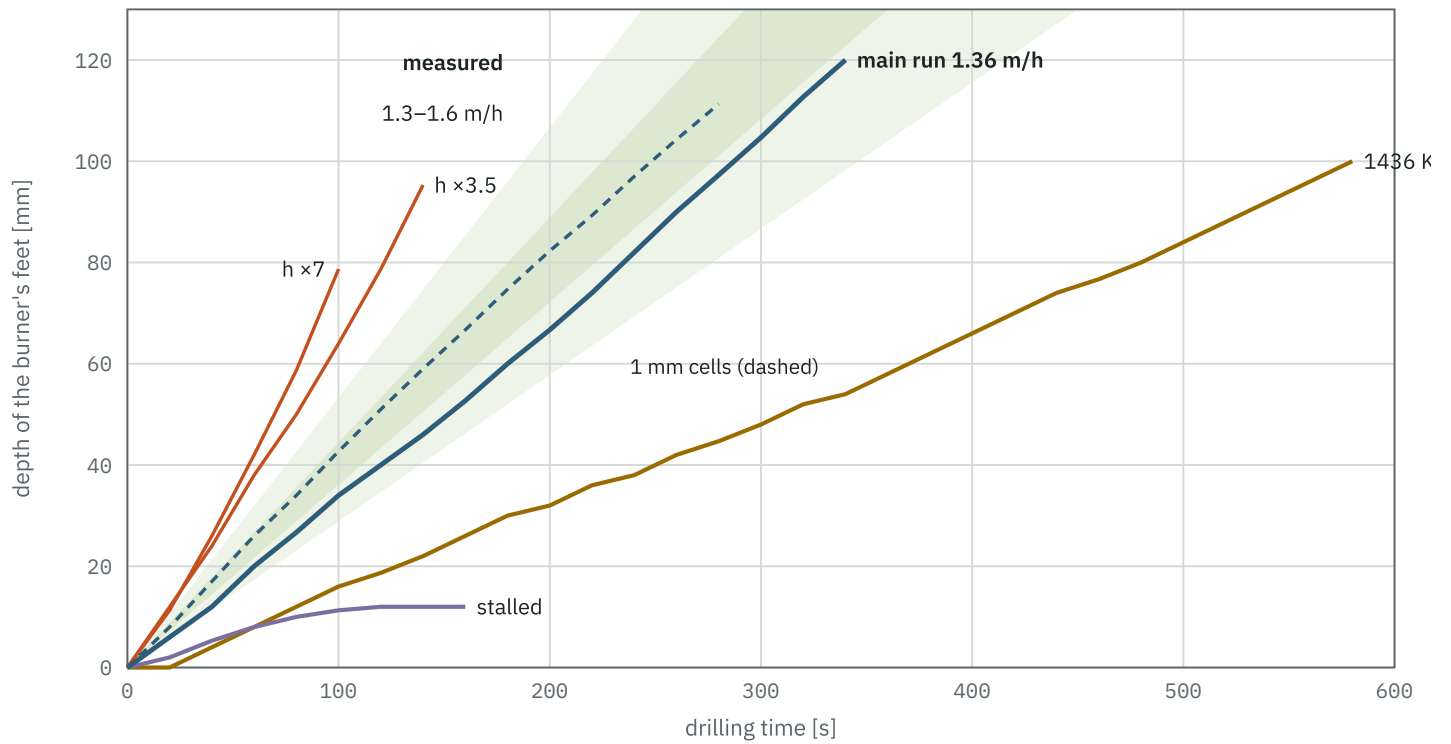
3. Results

The main run uses the correlation's heat transfer coefficient and the adiabatic flame temperature of 1900 K. It ran until the centre of the hole

reached the bottom of the simulated box, after 342 s of drilling and 0.12 m of burner descent.

Six runs are compared. Only the first is the prediction; the others show what each uncertain input does to it.

RUN	WHAT IS DIFFERENT	WHY IT IS RUN
Main run	Martin's coefficient at the measured flow (1449 W/m ² K), nozzle gas at 1900 K, jet entrains its own exhaust, 2 mm cells	The pre-registered prediction. This is the run that is scored.
1 mm cells	Same physics, half the cell size, box trimmed to 60 mm radius	Checks that the cell size does not set the answer.
h × 3.5 and h × 7	5 000 and 10 000 W/m ² K, the two literature values used before the correlation (Meier's Ch. 7 used 10 000, from Potter Drilling)	Shows how strongly the result depends on the least certain input, and whether the older values could have been right.
Gas at 1436 K	Nozzle gas at 1163 °C, the thermocouple's peak reading, instead of the adiabatic 1900 K	A low gas-temperature case. Not a measurement of the nozzle gas (see the table in §1).
Room-air mixing	The jet entrains 293 K air instead of its own exhaust	Tests the recirculation assumption of §2.5. The hand calculation predicted a stall for this case.



■ measured 1.3–1.6 m/h
 — main run (correlation h , 1900 K)
 - - - same, 1 mm cells
— literature $h \times 3.5$ and $\times 7$
— gas at 1436 K
— mixing with room air

Figure 4. How fast the burner sinks. Depth of the burner's feet against drilling time. The green wedge is the measured rate; the main run lies inside it and stays there. Runs end when the hole's centre reaches the bottom of the simulated box. The two literature values of the heat transfer coefficient drill too fast; the 1436 K gas drills too slowly; letting the jet mix with room air instead of its own exhaust makes the burner stall after 12 mm, as the hand calculation predicted.

CRITERION	MEASURED	MAIN RUN	VERDICT	COMMENT
Rate of penetration	1.3–1.6 m/h	1.36 m/h	pass	1.27–1.33 m/h with 1 mm cells. Still rising slowly at the end of the run.
Rock heating before a flake leaves	500–560 K	528 K	pass	
Narrowest hole diameter	≥ 80 mm	84.9 mm	pass	Still narrowing toward the burner's 80 mm at 0.5 m depth.
Hole diameter below 0.1 m depth	85–93 mm	92 → 85 mm	pass	From the wall profile, Figure 5.
Hole diameter averaged over the full 0.5 m	85–93 mm	104 mm	fail	Entirely due to a funnel in the top 0.1 m; see §4.
Volume removed, whole excavation	2.0–3.0 cm ³ /s	5.68 cm ³ /s	fail	2.3 times the measurement. Inside the 93 mm cylinder alone the model removes 3.27 cm ³ /s; the rest is the funnel. Meier's volume was measured by water filling, so the whole excavation is the like-for-like number.
Steady drilling state reached	yes (23 min)	no	open	The centre of the hole keeps deepening; see §4.
Cell-size check, 2 mm → 1 mm	within 5 %	–4.8 %	open	Within tolerance, but compared over a non-steady window.

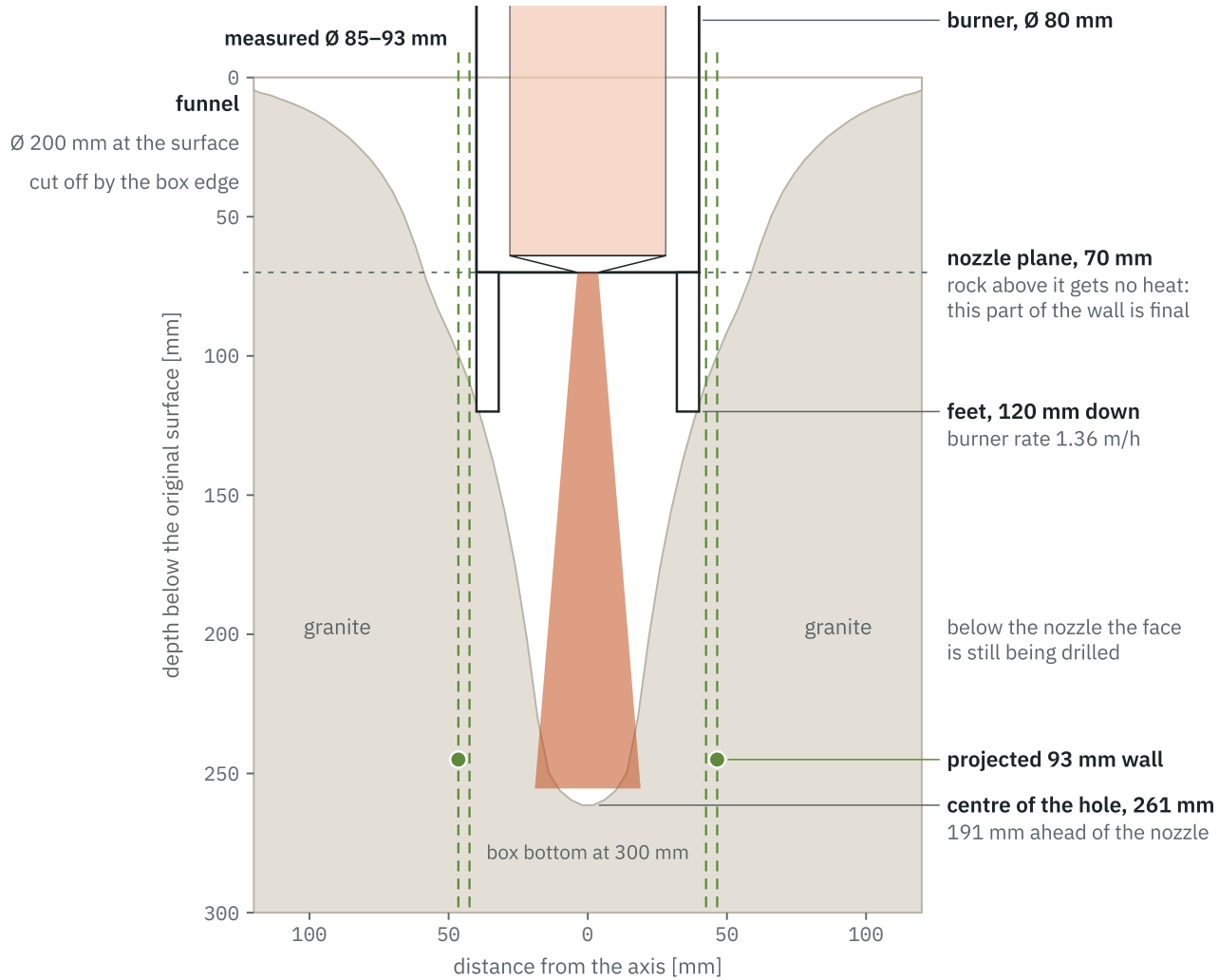


Figure 5. The simulated wellbore after 342 s of drilling. Vertical cross-section through the axis, to scale, from the main run. Grey is granite; the white region has been removed. The burner (outline) stands with its feet 120 mm down; its nozzle is 50 mm higher. Rock above the nozzle plane is out of the jet and no longer changes, so the wall above 70 mm depth is final: it flares to a 200 mm wide funnel at the surface. Below the nozzle the face is a deep bowl whose centre, 191 mm ahead of the nozzle, is still being drilled. The two dashed verticals mark the measured hole diameter; the marked point is where the model's own removal rates put the 93 mm wall once the burner has passed (245 mm). The simulated box is 120 mm wide from the axis, which truncates the funnel.

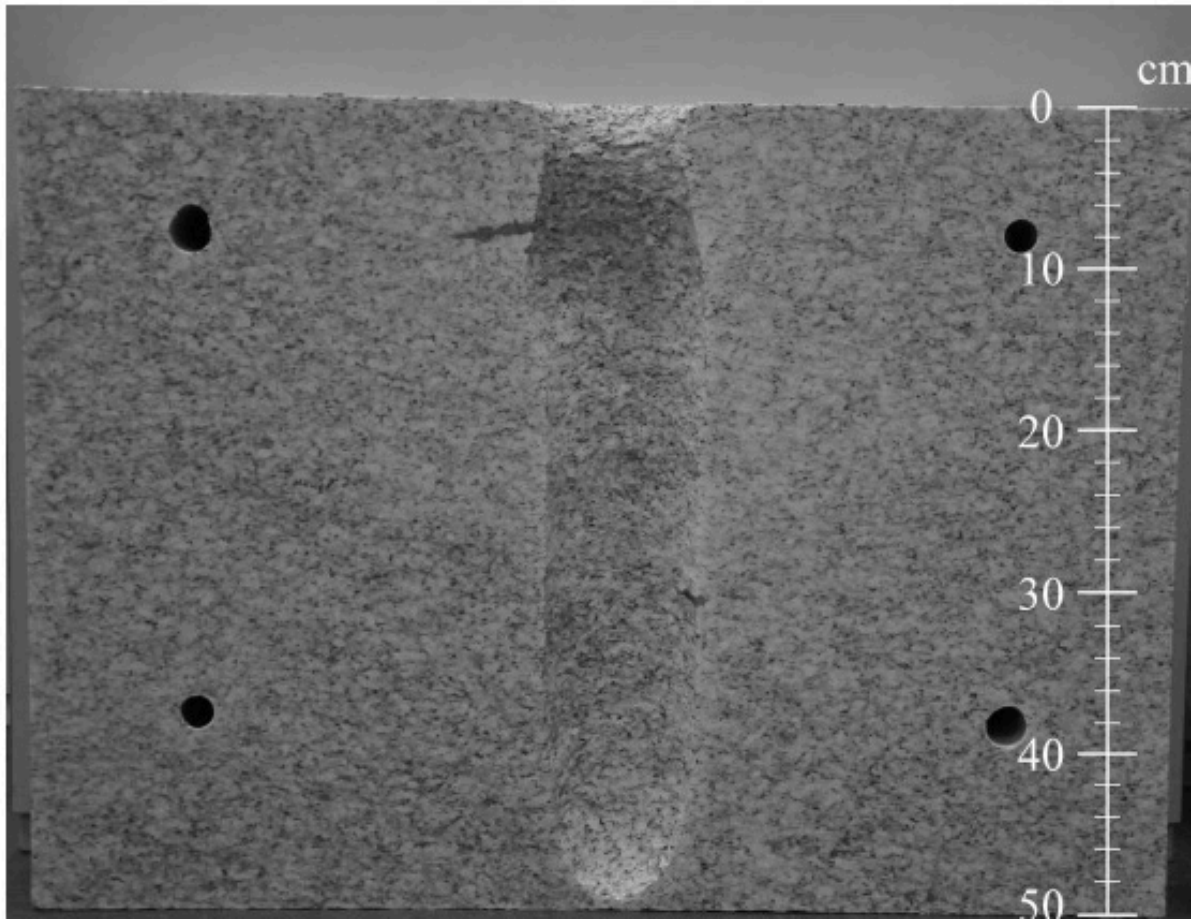


Figure 6. The hole Meier drilled (his Figure 8.8, p. 224): the 0.7×0.5 m block sawn through the hole, scale in centimetres on the right. The hole is a near-uniform channel from top to bottom, with a modest widening in the top few centimetres and a local underream at about 10 cm depth, which the thesis attributes to holding the burner still for a few seconds (p. 223). The lighter band beside the hole is thermally altered rock, not excavation. Compare Figure 5: the simulation matches this shape below about 10 cm, but above it opens into a 200 mm funnel that is not in the photograph.

Sensitivity: which unknown controls the answer

CHANGE FROM THE MAIN RUN	RATE OF PENETRATION	READING
Gas temperature 1900 K → 1436 K (thermocouple reading)	1.36 → 0.67 m/h	About 0.11 % per kelvin. A flame at 1700 K would sit on the lower edge of the accepted range. This is the dominant uncertainty.
Heat transfer coefficient ×3.5 / ×7 (the older guessed values)	2.43 / 2.82 m/h	Both too fast. The correlation value is the one that fits.
Jet mixes with room air instead of its own exhaust	stalls after 12 mm	The centre of the hole races ahead, the gas reaching the ring under the feet cools below the firing temperature, and the burner stops. Predicted by hand before the run.
Cell size 2 mm → 1 mm	−6 %	Not controlling.

The two unknowns, flame temperature and heat transfer coefficient, have plausible ranges that straddle the measurement, and the most defensible pair of values lands inside it without adjustment. That is a good position for a first-principles model. It also means the model cannot yet discriminate within the measured range: the pass depends on the flame being close to adiabatic.

4. What the two failures mean

The funnel at the mouth

In the top 0.1 m the simulated hole is far too wide: 216 mm at 10 mm depth, 132 mm at 50 mm, and only reaching the measured 92 mm at 100 mm depth. Below that it is right. The averaged-diameter and volume failures are this one feature.

The cause is in the gas model. Before a hole exists, the jet spreads over the free surface of the block. The model lets the gas lose heat only to the rock and never dilutes it with the surrounding air, so it is still at about 1270 K 170 mm from the axis and keeps removing rock there. Exhaust recirculation is also switched on from the first instant, before there is a hole to recirculate in.

This is a real error, not a scoring artefact. The sectioned block (Figure 6) shows a near-cylindrical hole to within a few centimetres of the surface, and the measured volume was obtained by filling the whole hole with water, so it would have included a funnel had there been one. The model removes 2.3 times that volume, almost all of it in the top 0.1 m.

No steady state

In every run the centre of the hole deepens faster than the ring under the feet and eventually reaches the bottom of the simulated box. The experiment drilled steadily for 23 minutes, so a real steady shape exists.

The cause is a limit of the heat transfer correlation. It is only valid for nozzle heights up to twelve nozzle diameters (90 mm), and beyond that the model simply holds the value constant. The centre of the face therefore keeps almost twice the ring's heating however far ahead it runs, and can never slow down to the ring's pace. This is an extrapolation choice, not physics.

Meier expected a deep centre himself: raising the power with the same nozzle, he wrote, would make the advancing face "look like a half elongated spheroid" (p. 222). A centre that runs ahead is plausible; a centre that never stops is the clamp. The ring, and hence the rate of penetration, feels the deep centre only through how much of the flame's heat it consumes. So the rate result survives; the claim to have reached the experiment's steady state does not, and the rate was still creeping upward when the runs ended.

5. Where the model stands

Three days before these runs, the same test was failing by a factor of four: the model then treated the flame as a fixed heat patch of guessed size, and drilled a narrow hole far too fast. Three changes brought it into range, each a piece of physics rather than a tuning: the correlation shape of the jet's

heating, the finite heat budget of the gas as it spreads, and the burner resting on its feet. The remaining disagreement is concentrated in the two items above. Meier's own outlook asked for this kind of model: "a failure criteria for the cleavage of a spall in order to analyze and model the rate of penetration of the drill. This requires additionally a moving boundary condition" (p. 250); his Chapter 7 model had neither and ran for ten seconds without removing rock.

Limitations to keep in mind

- **Transient scores.** Every number in §3 is taken 3 to 6 minutes into the run in a box 0.30 m deep, not from a steady state.
- **Two unmeasured inputs.** The nozzle gas temperature (1436–1900 K) and the heat transfer coefficient carry the result. Neither can be pinned down from the rock side.
- **The correlation is stretched.** At the end of the main run, 88 % of the face lies outside the range of jet heights and radii the correlation was fitted for.
- **The jet is supersonic; the correlation is not.** The Laval nozzle with its 6 mm throat is choked at this flow, so the jet leaves at supersonic speed from a chamber near 6 bar. Martin's data are for subsonic jets. The Reynolds number, set by the mass flow, is the same, but the impingement structure is not.
- **The nozzle gas temperature has no measurement behind it.** The only sensor sits upstream of the lifted flame (Figure 2). 1900 K is the adiabatic upper bound; the cooling-water heat load that would lower it is not reported in the chapter.
- **Side walls are not heated.** The rising exhaust warms the hole wall in reality; the model applies heat only to the face, on its horizontal projection, although the bowl reaches 80° slopes.
- **Rock under the feet is heated** as if the feet were not there.
- **Three feet are approximated** by a four-fold symmetric quarter model.
- **Flake size is not predicted.** Removal happens one 2 mm cell at a time; the thesis's particle-size data are not addressed.
- **The 1 MPa side clamp is not represented.** The thesis argues it has little effect at this scale.

Next steps, in order of leverage

1. **Jet heating beyond the correlation's range.** Let the heating at the centre fall off with distance as free and confined jets do, so the centre of the hole settles and a steady state can be reached and scored. Also a taller simulated box.
2. **Mixing with the surrounding air near the surface.** Dilute the spreading gas with ambient air while the hole is shallow and switch to exhaust recirculation as it deepens. This is what should remove the funnel, if the thesis shows there was none.
3. **A nozzle-exit temperature from an energy balance of the burner** (fuel and air flows, dissociation, heat lost to the cooling water). This is the dominant uncertainty and cannot be settled by rock physics. The chapter gives the cooling-water flow, 160 L/h, but not its temperature rise; the author or the design documents would have to supply it. Meier used exactly this kind of balance, cooling water included, to estimate jet temperatures in his pressure-vessel burner (thesis §4.2, pp. 119–123).
4. **Shield the rock under the feet from the jet.**
5. **Heat the side walls** with the rising exhaust and account for the true slope of the face.

Provenance. Experiment: Meier 2017, ETH Diss. 24021, Ch. 8, and burner drawing set VT5. Jet heating: H. Martin, *Advances in Heat Transfer* 13 (1977). Granite properties: thesis Table 7.1. Firing temperature: crack-growth criterion with microcrack lengths of a few micrometres, drawn from a Weibull distribution. Simulation records: `tests/MMWSpalling/studies/d2b_feet_rop/RESULTS.md` and the run outputs there (runs `B_JM_A2`, `B_J5_A2`, `B_J10_A2`, `C_1436_A2`, `C_fixed_A2`, `D_1mm_A2`); Figures 4 and 5 are drawn directly from those outputs; Figures 2 and 6 are reproduced from the thesis (its Figures 8.7 and 8.8, pp. 220 and 224). Reviewer's notes: `Claude_markdowns/2026-09-16.md`, `2026-09-16b.md`, and the thesis reading notes `Claude_markdowns/meier_thesis_notes.md`.